

Cadillac's WWTP- a short History

The City of Cadillac WWTP is an advanced treatment plant providing treatment to the City of Cadillac and to several surrounding townships. The daily flow for 2011 averaged 2.14 MGD. The treatment process is comprised of grit removal, primary treatment, secondary treatment (activated sludge), RBC treatment for ammonia removal, Cloth media filtration and Ultra Violet (UV) Disinfection. The effluent is discharged into the Clam River which is designated as a cold water trout stream several hundred yards from our discharge.

All solids are removed from the primary tanks to the anaerobic digester. From the primary digester solids are transferred to a secondary covered digester which is not heated. This tank is used for solids separation and supernatant withdrawal. Solids from this tank are pumped to one of two storage tanks. Two times a year the city contracts with a sludge application firm for solids injection on farm lands.

Cadillac Has treated sewage for over one hundred. In 1908 the first plant was built in Cadillac that consisted of Five filter beds that used slag for filter media.

In 1923 a new plant was built. This facility utilized a septic tank with a capacity of 444,312 gallons, a 5600 sq. Ft sludge bed and contact filter with 118,272 cu. Ft. Of slag filter available. In 1939 5 ½ acres of sand filters operated in series was added.

In 1962 a new "Activated Sludge Treatment Plant" was built. With expansions in 1975, 1989 and 1996 the current plant was established with upgrades in 2003 and 2007.

The City of Cadillac looks forward to the future challenges that will occur in the wastewater field. Cadillac has always had an interest in providing the best treatment process for its wastewater and has tried to do this for the past 95 years. With any luck Cadillac will be able to continue its stewardship of the public's water for years to come.

Cadillac WWTP

1995-1996 Expansion

Primary clarification was added (0.17 MG)
Biosolids storage was added (0.7 MG)
Secondary clarification was added (0.09 MG)
Fine bubble aeration was added (0.33 MG)

2003 Upgrade:

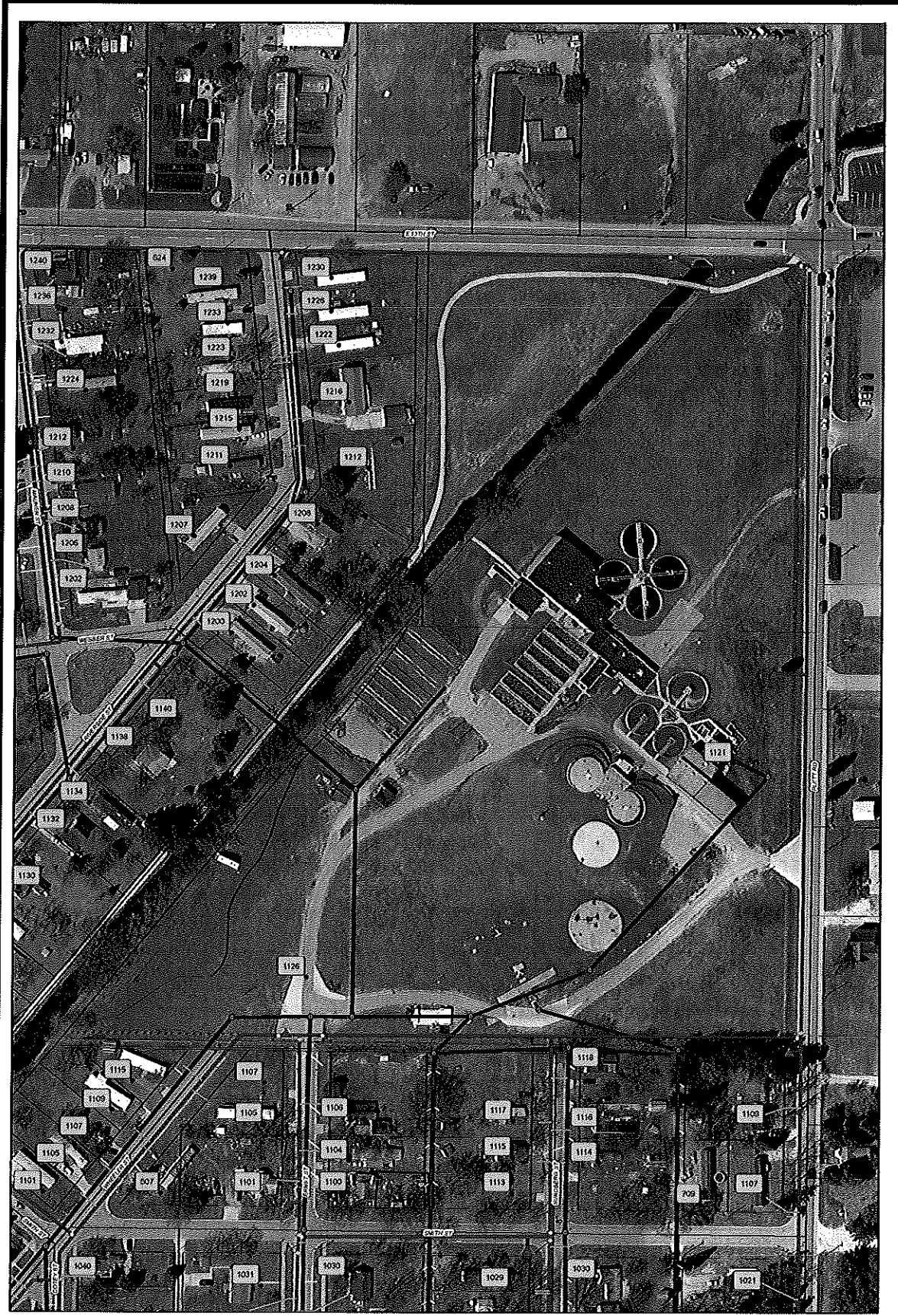
Replaced chlorine disinfection with U.V. disinfection

2006-2008 Upgrades:

Sand filters replaced w/ cloth media (Aqua Disk)
Additional U.V. banks installed
Secondary screw pumps replaced
Digester retrofitted with chopper pump
Digester gas holding cover and floating cover replaced
Biosolid tank #3 was retrofitted

Lift Stations:

Leeson – new Pumps, Controls, and generator were installed
Lake – new Pumps, Controls, generator, and new force main were installed
Leisure – new Pumps, Controls, and generator were installed

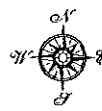


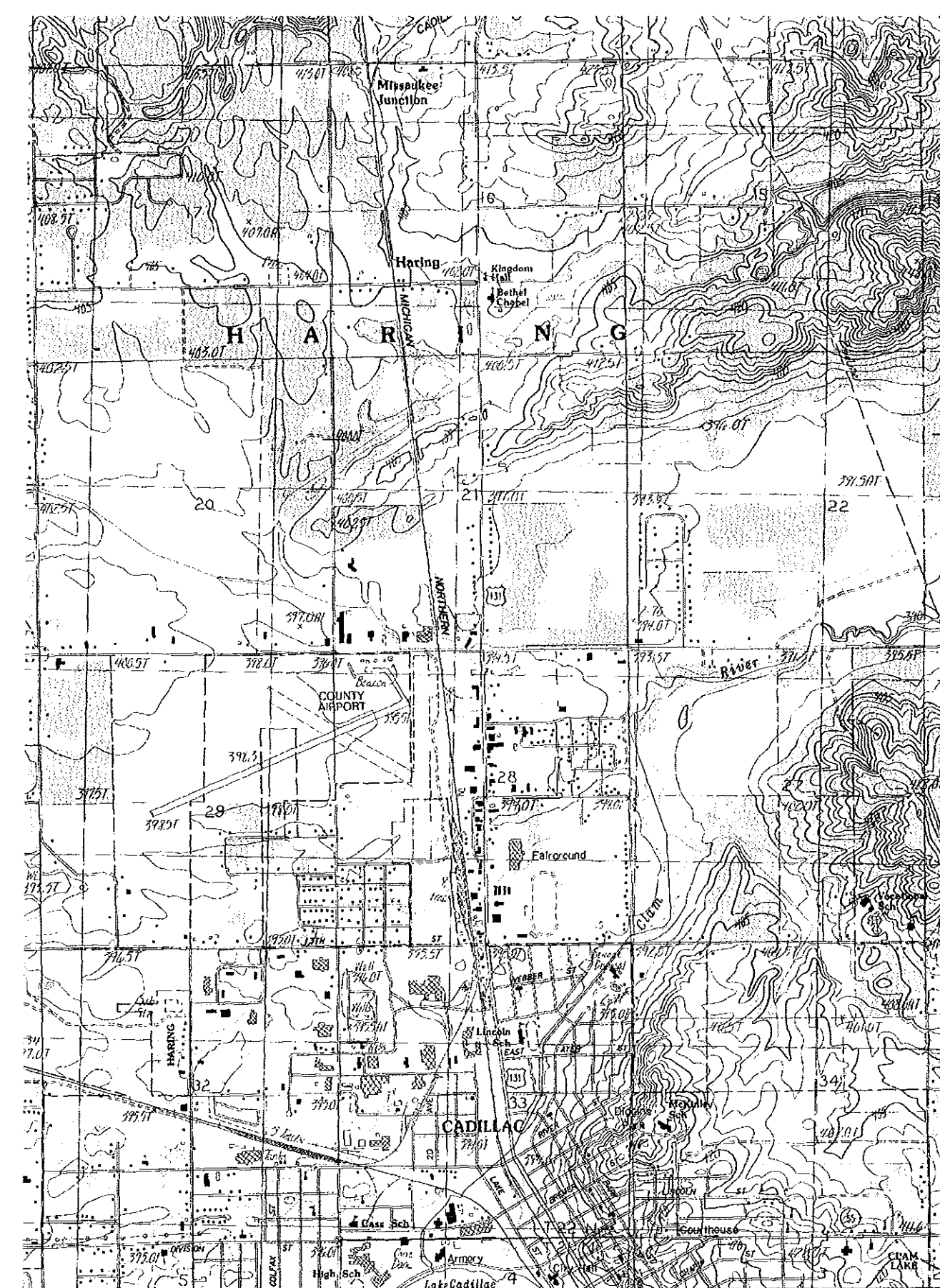
Prepared By:
Utilities Department
City of Cadillac
Cadillac, MI 49601
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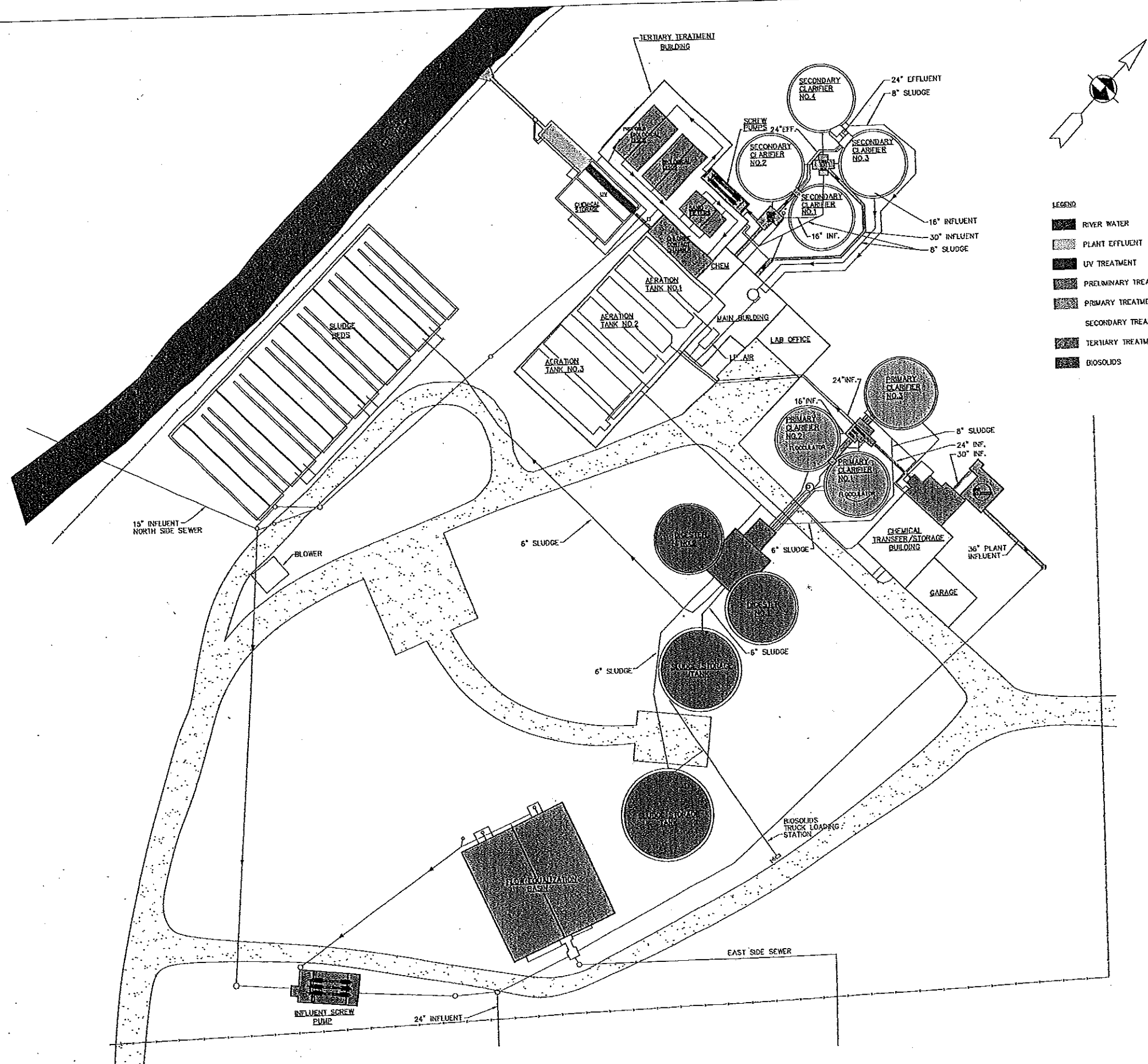
Date of Aerials
April 7, 2005

0 62.5 125 250 375 500 Feet

CITY OF
CADILLAC
MICHIGAN







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MICHIGAN
WASTEWATER TREATMENT PLANT

SITE PLAN

Ayres, Lewis, Norris & May, Inc.
Ann Arbor, Michigan